Year 1 AS Statistics and Mechanics

Chapter 7

Hypothesis testing

Practice Booklet

Contents

7.1 Hypothesis testing ..... book p. 45
7.2 Finding critical values ..... book p. 46
7.3 One-tailed tests ..... book p. 47
7.4 Two-tailed tests ..... book p. 48
Problem solving: Set A ..... book p. 50
Problem solving: Set B ..... book p. 51
Selected exam question bank questions .....
Answers .....

7.1 Hypothesis testing

Practice Book, printed page 45.

1. Alice is testing whether a four-sided dice is biased towards the number three. She rolls the dice 50 times and counts the number of threes she gets.

a. Describe the test statistic.

b. Write down a suitable null hypothesis.

c. Write down a suitable alternative hypothesis.

Hint

The null hypothesis is the hypothesis that you assume to be correct when making your calculations.

2. For each of these hypotheses, state whether they describe a one-tailed or a two-tailed test.

a. $H_0 : p = 0.7$, $H_1 : p < 0.7$

b. $H_0 : p = 0.5$, $H_1 : p > 0.5$

c. $H_0 : p = 0.4$, $H_1 : p \neq 0.4$

Hint

To test whether p is different from a given value you need to use a two-tailed test.

3. E/P A Member of Parliament has found that, from long experience, he gets support from 65% of the adult population of his constituency. A new sample of 50 people is taken and 24 people say that they support him. The Member of Parliament wants to test, at the 5% level of significance, whether or not his support has decreased.

a. Suggest a suitable test statistic. (1 mark)

b. Write down two suitable hypotheses. (2 marks)

c. Explain the condition under which the null hypothesis is rejected. (2 marks)

4. E/P A cereal manufacturer places a prize in some packets of a type of cereal. The manufacturer claims that there are prizes in 5% of packets. A regular customer purchases 120 packets of the cereal and finds a prize in 15 packets. She wishes to test, at the 10% level of significance, whether the manufacturer is putting prizes in more than 5% of the packets.

a. Write down a suitable test statistic. (1 mark)

b. Write down suitable null and alternative hypotheses. (2 marks)

c. Explain the condition under which the null hypothesis would be accepted. (2 marks)

5. E/P Georgia runs a car-servicing workshop. She claims that 60% of all services take less than two hours to complete. Gemma, her business partner, thinks that this proportion might be incorrect. She takes a sample of 50 services and finds that five take longer than two hours to complete. She wishes to test, at the 5% significance level, whether Georgia's claim is incorrect.

a. Write down a suitable test statistic. (1 mark)

b. Write down suitable null and alternative hypotheses. (2 marks)

c. Explain the condition under which the null hypothesis is rejected. (2 marks)

7.2 Finding critical values

Practice Book, printed page 46.

1. A test statistic has a distribution $\mathrm{B}(8, p)$. Given that $\mathrm{H}_0 : p = 0.6$ and $\mathrm{H}_1 : p < 0.6$,

a. find the critical region for the test using a 5% level of significance

b. write down the actual significance level of the test.

Hint

Assume that H_0 is true, so that $X \sim \mathrm{B}(8, 0.6)$. You need to find the largest value of x such that $\mathrm{P}(X \leq x) < 0.05$.

2. A test statistic has a distribution $\mathrm{B}(15, p)$. Given that $\mathrm{H}_0 : p = 0.3$ and $\mathrm{H}_1 : p > 0.3$,

a. find the critical region for the test using a 10% level of significance

b. write down the actual significance level of the test.

Hint

The actual significance level of the test is the probability that the test statistic is greater than or equal to the critical value. It is the probability of incorrectly rejecting the null hypothesis.

3. A test statistic has distribution $\mathrm{B}(25, p)$. Given that $\mathrm{H}_0 : p = 0.42$ and $\mathrm{H}_1 : p \neq 0.42$,

a. find the critical region for the test using a 10% level of significance

b. write down the actual significance level of the test.

Hint

This is a two-tailed test so the critical region will have two 'tails', one at each end of the distribution.

4. E/P A florist claims that 20% of hydrangeas will have blue flowers. A regular customer thinks that this proportion is overstated and decides to buy 30 plants in order to test the claim.

a. Describe the test statistic and write down suitable null and alternative hypotheses. (3 marks)

b. Using a 5% level of significance, find the critical region for the customer's test. (2 marks)

c. State the probability of incorrectly rejecting the null hypothesis using this critical region. (1 mark)

5. E/P A hotel typically gets 5-star reviews 60% of the time on a travel website. The hotel owner decides to make some improvements to try and increase this percentage. The owner takes a random sample of 20 reviews after the improvements have taken place.

a. Describe the test statistic and write down suitable null and alternative hypotheses. (3 marks)

b. Using a 10% level of significance, find the critical region for the hotel owner's test. (2 marks)

c. State the actual significance level of the test. (1 mark)

6. E/P A manufacturer of electronic components finds that, over the long term, one component in every 20 of the components produced has a fault. The manufacturer decides to modify the production process and believes that this will change the proportion of components that have a fault.

a. Suggest a suitable model and write down suitable null and alternative hypotheses. (3 marks)

The manufacturer takes a random sample of 50 components.

b. Find, at the 20% level of significance, the critical region for a test to check the belief that the proportion of components with a fault has changed. (4 marks)

c. State the actual significance level of the test. (1 mark)

7. E/P A random variable has distribution $\mathrm{B}(25, p)$. A single observation is used to test $\mathrm{H}_0 : p = 0.3$ against $\mathrm{H}_1 : p \neq 0.3$.

a. Using a 5% level of significance, find the critical region for this test. The probability in each tail should be as close as possible to 0.025. (3 marks)

b. Write down the actual significance level of the test. (2 marks)

7.3 One-tailed tests

Practice Book, printed page 47.

1. A single observation, x, is taken from a binomial distribution $\mathrm{B}(20, p)$ and a value of 3 is obtained. Use this observation to test $\mathrm{H}_0 : p = 0.4$ against $\mathrm{H}_1 : p < 0.4$ using a 2% level of significance.

Hint

Assume $X \sim \mathrm{B}(20, 0.4)$ and calculate $\mathrm{P}(X \leq 3)$, and compare this with 0.02.

2. A random variable X has distribution $\mathrm{B}(25, p)$. A single observation of $x = 17$ is taken from the distribution. Test, at the 1% level of significance, $\mathrm{H}_0 : p = 0.45$ against $\mathrm{H}_1 : p > 0.45$.

Hint

You could find $\mathrm{P}(X \geq 17)$, or you could start by finding the critical region for the test.

3. E A magician claims to be able to fool a person by sleight of hand 90% of the time. In a sample of n people, X is the number of people who are fooled.

a. Write down a suitable distribution to model X. (1 mark)

A random sample of 40 people is taken and it is found that 33 are fooled by the sleight of hand. It is believed that the magician's claim of fooling people 90% of the time is overstated.

b. Test this claim at the 5% level of significance. State your hypotheses clearly. (5 marks)

4. E/P Tia grows marrows and she finds that over time, 60% of the marrows are longer than 40 cm. Tia uses a new fertiliser. The manufacturer claims the fertiliser will increase the proportion of marrows that are longer than 40 cm.

Tia takes a random sample of 30 marrows and carries out a hypothesis test, at the 5% level of significance, to test the manufacturer's claim.

a. Write down Tia's null and alternative hypotheses. (2 marks)

b. Find the critical region for Tia's test. (3 marks)

Tia finds that 25 marrows in her sample are longer than 40 cm.

c. Comment on this observation in light of the critical region. (2 marks)

5. E/P Esha makes clay pots. She estimates that 10% of the pots fracture during the firing process in her kiln. She decides to change the type of clay she uses in order to try and reduce the proportion of pots that fracture. She takes a random sample of 50 pots made using the new clay and finds that only one has fractured in the kiln.

Stating your hypotheses clearly and using a 5% level of significance, test whether there is evidence that changing the type of clay has reduced the likelihood of the pots fracturing. (5 marks)

6. E/P A random variable has distribution $B(10, p)$. A single observation is taken and a value of 7 obtained.
Briony decides to test $H_0 : p = 0.3$ against $H_1 : p > 0.3$, using a 1% level of significance.
Her working is shown below.

> Let $X \sim B(10, 0.3)$
> $P(X = 7) = 0.0090$
> Since $0.0090 < 0.01$, reject H_0
> There is evidence that $p > 0.3$

Briony has made a mistake in her working.
Explain her mistake and determine the actual result of the test. (3 marks)

7. E/P From experience, a door-to-door carpet salesman knows that the probability that he makes a sale is 0.05. He visits 15 houses in one day.

a. Calculate the probability that he makes at least one sale. (2 marks)

He decides to change his pitch to try and improve the likelihood of making a sale. On the day after changing his pitch, he visits 26 houses and makes 4 sales.

b. Test, at the 5% level of significance, the claim that he has increased his likelihood of making a sale. State your hypotheses clearly. (5 marks)

7.4 Two-tailed tests

Practice Book, printed page 48.

1. A single observation, x, is taken from a binomial distribution $X \sim B(15, p)$ and a value of 3 is obtained.
Use this observation to test, at the 10% level of significance, $H_0 : p = 0.4$ against $H_1 : p \neq 0.4$.

Hint

Halve the significance level in each tail. You need to work out $P(X \leq 3)$ and compare the answer with 0.05.

2. A random variable has distribution $X \sim B(40, p)$. A single observation of $x = 34$ is taken from the distribution. Test, at the 5% level of significance, $H_0 : p = 0.7$ against $H_1 : p \neq 0.7$.

Hint

Find the probability that X is greater than or equal to 34 given the value of p.

3. E/P The proportion of people diagnosed with a particular disease nationally is 0.05. A random sample of 200 people is taken from the south west. It is found that 16 people have the disease. Test, at the 5% level of significance, the hypothesis that the proportion of people in the south west who have the disease is different from the national average. State your hypotheses clearly. (5 marks)

4. E/P A nutritionist finds that, on average, 10% of her clients are overweight. She decides to introduce a new diet to her clients.
She takes a random sample of 50 clients who have used the new diet.

a. Find the critical regions, at the 10% level of significance, to test whether or not the proportion of her clients has changed. (4 marks)

b. State the actual significance level of the test. (1 mark)

From the sample, she finds that 2 clients are overweight.

c. Comment on this finding in light of your critical regions. (2 marks)

5. E/P Meghan rolls an eight-sided dice 120 times and finds that it lands on the number eight 25 times. Use a two-tailed test with a significance level of 5% to determine whether there is sufficient evidence to conclude that the dice is biased. State your hypotheses clearly. (5 marks)

6. E/P A single observation, x, is taken from a binomial distribution $\mathrm{B}(30, p)$. This observation is used to test $\mathrm{H}_0 : p = 0.25$ against $\mathrm{H}_1 : p \neq 0.25$.

a. Using a 10% level of significance, find the critical region for this test. The probability in each tail should be as close as possible to 5%. (3 marks)

b. State the actual significance level of this test. (2 marks)

The actual value of x obtained is 11.

c. State a conclusion that can be drawn based on this value, giving a reason for your answer. (1 mark)

7. E/P A chicken farmer knows from experience that one egg in every 50 has a double yolk. The farmer decides to change the type of feed that she uses. She wants to carry out a hypothesis test to determine whether the new feed has affected the chance of a double yolk.
In a random sample of 50 eggs produced by chickens using the new feed, she finds that 3 have double yolks.
Carry out a hypothesis test, at the 10% level of significance, to assess whether the chance of a double yolk has changed. State your hypotheses clearly. (5 marks)

Problem solving: Set A

Bronze

A single observation, x, is taken from a binomial distribution $\mathrm{B}(30, p)$, and is used to test $\mathrm{H}_0 : p = 0.36$ against $\mathrm{H}_1 : p \neq 0.36$.

a. Find the critical region for a two-tailed test at the 5% level of significance. (3 marks)

b. Write down the actual significance level of the test. (1 mark)

The value $x = 16$ is observed.

c. Comment on this observation in light of your critical region. (1 mark)

Silver

A single observation, x, is taken from a binomial distribution $\mathrm{B}(35, p)$, and is used to test $\mathrm{H}_0 : p = 0.45$ against $\mathrm{H}_1 : p \neq 0.45$.

a. Find the critical region for a two-tailed test at the 10% level of significance. The probability in each tail should be as close as possible to 0.05. (3 marks)

b. Write down the probability of incorrectly rejecting the null hypothesis in this test. (1 mark)

The value $x = 22$ is observed.

c. Comment on this observation in light of your critical region. (1 mark)

Gold

The random variable X is known to have the distribution $\mathrm{B}(20, p)$, where p is unknown. Amy uses a single observation from X to test $\mathrm{H}_0 : p = 0.4$ against $\mathrm{H}_1 : p \neq 0.4$.

a. Find the critical region for Amy's test at the 5% level of significance. (3 marks)

Amy decides she will carry out her test 15 times. She will reject H_0 if more than a quarter of her observations lie in the critical region.

b. Find the probability that Amy will incorrectly reject H_0. (3 marks)

Problem solving: Set B

Bronze

A juggler claims to be able to juggle five balls for over ten minutes 65% of the times she attempts it. A circus owner believes that this proportion is not correct. The juggler attempts to juggle five balls 15 times and juggles for over ten minutes on 13 occasions.
Test, at the 10% level of significance, whether this provides evidence for the circus owner's belief. State your hypotheses clearly. (5 marks)

Silver

The outcome of rolling a biased four-sided dice is modelled as a discrete random variable, X, with probability mass function $\mathrm{P}(X = x) = \dfrac{x}{k}, \quad x = 1, 2, 3, 4$

a. Calculate $\mathrm{P}(X = 4)$. (3 marks)

Doreen claims that the model is incorrect and rolls the dice 50 times. She gets a 4 on 28 occasions.

b. Using a 5% level of significance and stating your hypotheses clearly, test Doreen's claim. (4 marks)

Gold

A paper company finds that, over time, defects occur at a rate of 1 in every 250 sheets of paper produced. After switching to recycled paper, the company examines a sample of 300 sheets of paper, and finds that it contains 5 defective sheets. An employee decides to carry out a hypothesis test, at the 5% level of significance, to assess the claim that the switch to recycled paper has affected the rate of defects. The employee's working is shown below:

> Let X be the random variable 'number of defects' and $X \sim \mathrm{B}(300, p)$
> $\mathrm{H}_0 : p = 0.004$
> $\mathrm{H}_1 : p \neq 0.004$
> $\mathrm{P}(X \leq 5 \mid p = 0.004) = 0.9985$
> Since $0.9985 > 0.05$, accept H_0
> There is no evidence that the rate of defects has changed.

The employee has made two mistakes.
Explain these mistakes and determine the correct conclusion to the hypothesis test. (5 marks)

Now try this → Exam question bank Q23, Q27, Q29, Q31, Q34

Selected exam question bank questions

Question 23

The traffic police in a town believe that 20% of impounded cars have illegal tyres. A random sample of 30 cars from an impound lot is taken.

a. Give two reasons to support the use of a binomial distribution to model the random variable X, the number of cars in the sample that have illegal tyres. (2 marks)

b. Calculate the probability that the sample contains fewer than 4 cars with illegal tyres. (2 marks)

The Department for Transport disagrees with the belief of the police and claims that the percentage of impounded cars with illegal tyres is less than 20%.

c. Write down suitable null and alternative hypotheses that can be used to test this claim. (2 marks)

The Department for Transport tests its claim using a 10% level of significance.

d. Given that there were 3 cars in the sample that had illegal tyres, comment on the Department for Transport's claim. (2 marks)

Question 27

The head of sixth form suggests that the probability that Savannah is late to school is 0.3. Savannah does not believe this and, to test the claim, she records the number of days she is late over a random sample of 25 days.

a. Define the term 'critical region' in the context of hypothesis testing. (1 mark)

b. Find the critical region for a two-tailed test, at the 5% level of significance, to test whether the probability Savannah is late to school is 0.3. The probability of rejecting each tail should be as close as possible to 2.5%. (4 marks)

c. State the actual significance level of the test. (1 mark)

From the sample, Savannah finds that she was late on 4 occasions.

d. Comment on whether Savannah should accept the head of sixth form's claim in light of this evidence. (2 marks)

Question 29

A random variable X has distribution $\mathrm{B}(15, p)$.

Amy uses a single observation of X to test $\mathrm{H}_0 : p = 0.34$ against $\mathrm{H}_1 : p \neq 0.34$.

a. Using a 10% level of significance, find the critical region for Amy's test. The probability in each tail should be as close as possible to 0.05. (3 marks)

b. Write down the actual significance level of Amy's test. (2 marks)

Donna decides to take two independent observations of X and reject H_0 only if both of her observations fall within the critical region.

c. State, with a reason, whether the actual significance level of Donna's test is larger or smaller than Amy's. (1 mark)

Question 31

Charlie is playing a game with her friend Mikhela. The probability of her winning any one game is 0.2. Charlie and Mikhela play the game 25 times.

a. Find the probability that Charlie wins exactly 6 of the games. (1 mark)

b. Find the probability that Charlie wins more than 8 games. (2 marks)

Charlie claims that she has a trick that she can use to increase her probability of winning. To test the claim, Charlie and Mikhela play the game 25 more times and Charlie wins 9 times.

c. Write down suitable null and alternative hypotheses for this test. (2 marks)

d. Using a significance level of 5% and your answer to part **b**, state the outcome of the test. You must justify your answer. (2 marks)

Question 34

From the large data set, the probability of a randomly chosen day in Camborne in 1987 having 7 or 8 oktas of cloud cover is 0.36.

Maedeh believes the likelihood getting 7 or 8 oktas of cloud cover increased in 2015. She decides to take a random sample of 42 days from the large data set for Camborne in 2015, stratified by month.

a. Explain briefly how Maedeh could take her sample. (2 marks)

b. State what is meant by a null hypothesis. (1 mark)

c. Write down two suitable hypotheses that Maedeh can use to test her belief. (1 mark)

From her sample, she finds that there are 7 or 8 oktas of cloud cover on 22 days.

d. Using a 5% level of significance, test whether or not there is evidence supporting Maedeh's belief. (4 marks)

Answers

7.1 Hypothesis testing

1.

a. The number of threes Alice rolls.

b. $H_0 : p = 0.25$

c. $H_1 : p > 0.25$

2.

a. One

b. One

c. Two

3.

a. The number of people in the sample who say they support the MP.

b. $H_0 : p = 0.65$, $H_1 : p < 0.65$

c. If the probability of 24 or fewer people in the sample supporting the MP (with $p = 0.65$) is less than or equal to the significance level (5%).

4.

a. The number of packets of cereal in the sample that contain a prize.

b. $H_0 : p = 0.05$, $H_1 : p > 0.05$

c. If the probability of 15 or more packets containing a prize (with $p = 0.05$) is greater than the significance level (10%).

5.

a. The number of cars in the sample that take less than or equal to 2 hours to complete the service.

b. $H_0 : p = 0.6$, $H_1 : p \neq 0.6$

c. If the probability of 45 or more services taking less than two hours (with $p = 0.6$) is less than half the significance level.

7.2 Finding critical values

1.

a. $0 \leq X \leq 2$

b. 4.98%

2.

a. $8 \leq X \leq 15$

b. 5.00%

3.

a. $0 \leq X \leq 6$ and $16 \leq X \leq 25$

b. 7.15%

4.

a. The number of plants in the sample that have blue flowers. $H_0 : p = 0.2$, $H_1 : p < 0.2$

b. $0 \leq X \leq 2$

c. 4.42%

5.

a. The number of 5-star reviews in the sample. $H_0 : p = 0.6$, $H_1 : p > 0.6$

b. $16 \leq X \leq 20$

c. 5.10%

6.

a. $X \sim B(n, 0.05)$; $H_0 : p = 0.05$, $H_1 : p \neq 0.05$

b. 0 and $5 \leq X \leq 50$

c. 18.1%

7.

a. $\{0 \leq X \leq 3\}$ and $\{13 \leq X \leq 50\}$

b. 5.07%

7.3 One-tailed tests

1. Reject H_0 (prob = 0.016)

2. Accept H_0 (prob = 0.017)

3.

a. $X \sim B(n, 0.9)$

b. $H_0 : p = 0.9$, $H_1 : p < 0.9$. Accept H_0 (prob = 0.0995)

4.

a. $H_0 : p = 0.6$, $H_1 : p > 0.6$

b. 23 or more

c. 25 is in the critical region so reject the null hypothesis; there is evidence to support the manufacturer's claim.

5. $H_0 : p = 0.1$, $H_1 : p < 0.1$; probability of 'one or no pots fracturing' given $p = 0.1$ is 0.0338 which is less than 5%, so reject the null hypothesis; there is evidence that the new clay has reduced the likelihood of the pot fracturing.

6. Briony has calculated $P(X = 7)$ rather than $P(X \geq 7)$. $P(X \geq 7) = 0.0106 > 0.01$, so accept H_0.

7.

a. 0.5367

b. $H_0 : p = 0.05$; $H_1 : p > 0.05$. Given $X \sim B(26, 0.05)$, $P(X \geq 4) = 0.0387 < 0.05$ so reject H_0; there is evidence the salesman has increased his likelihood of making a sale.

7.4 Two-tailed tests

1. Accept H_0 (prob = 0.0905)

2. Reject H_0 (prob = 0.0238)

3. $H_0 : p = 0.05$, $H_1 : p \neq 0.05$; accept H_0 since probability of 16 or more people having the disease is $0.0444 > 0.025$. The proportion of people who have the disease in the south west is not different from the national average.

4.

a. $X \leq 1$ and $X \geq 10$

b. 5.83%

c. 2 is in the acceptance region, so no evidence to suggest the new diet has changed the proportion of clients who are overweight.

5. $H_0 : p = 0.125$, $H_1 : p \neq 0.125$; probability of 25 or more 8s given $p = 0.125$ is $0.0068 < 0.025$, so evidence that the dice is biased.

6.

a. $X \leq 3$ or $X \geq 12$

b. 8.81%

c. 11 lies in the acceptance region so accept H_0.

7. $H_0 : p = 0.02$; $H_1 : p \neq 0.02$. $P(X \geq 3) = 0.0784 > 0.05$ so accept H_0. There is no evidence to suggest that the probability of getting a double yolk is different.

Problem solving: Set A

Bronze

a. $X \leq 5$ or $X \geq 17$

b. 3.46%

c. 16 lies within the acceptance region so accept the null hypothesis.

Silver

a. $X \leq 10$ or $X \geq 21$

b. 0.08909

c. 22 lies within the critical region so reject the null hypothesis.

Gold

a. $X \leq 3$ or $X \geq 13$

b. 0.00184

Problem solving: Set B

Bronze $H_0 : p = 0.65$, $H_1 : p \neq 0.65$. Probability of 13 or more given $p = 0.65$ is 0.0617 which is greater than 0.05 so no evidence to support the circus owner's belief.

Silver

a. 0.4

b. $H_0 : p = 0.4$, $H_1 : p \neq 0.4$. Probability of 28 or more 4s given $p = 0.4$ is 0.0160 which is less than 0.025 so reject H_0; there is evidence to support Doreen's claim.

Gold The employee has calculated the probability in the wrong tail: $P(X \geq 5)$ should be calculated. The employee has not halved the significance level in making a comparison with the probability. $P(X \geq 5 \mid p = 0.004) = 0.0076 < 0.025$ so reject H_0; there is evidence that the rate of defects has changed.

Selected exam question bank answers

Question 23

a. A fixed probability, p, of success (0.2) and two possible outcomes (illegal and legal).

b. 0.1227

c. $H_0 : p = 0.2$, $H_1 : p < 0.2$

d. P(3 or fewer) $= 0.1227 > 0.10$, so there is insufficient evidence to support the Department of Transport's claim.

Question 27

a. A region of a probability distribution which, if the test statistic is within it, would cause you to reject the null hypothesis.

b. $X \leq 3$ and $X \geq 13$

c. 5.07%

d. Savannah should accept it since 4 lies in the acceptance region. There is evidence that she is late to school with probability 0.3

Question 29

a. $X \leq 2$ or $X \geq 9$

b. 10.7% (3 s.f.)

c. Smaller: Donna has a 1.14% chance of incorrectly rejecting the null hypothesis.

Question 31

a. 0.1633

b. 0.0468

c. $H_0 : p = 0.2$, $H_1 : p > 0.2$

d. Critical region is 9 or more (with a significance level of 5%) and since 9 lies in the critical region there is evidence to support Charlie's claim.

Question 34

a. Choose 7 days at random from each of the 6 months in the data set.

b. The hypothesis that you assume to be correct.

c. $H_0 : p = 0.36$, $H_1 : p > 0.36$

d. Probability of '22 or more' given $p = 0.36$ is 0.0218, which is less than 5% so there is evidence to support Maedeh's belief.